

Assignment Code: DA-AG-006

Statistics Advanced - 1 | Assignment

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 200

Question 1: What is a random variable in probability theory?

Answer:

Random Variable in Probability Theory

Definition:

A random variable is a numerical quantity whose value is determined by the outcome of a random experiment. In other words, it is a function that assigns a real number to each possible outcome in the sample space of a random process.

Formally, if S is the sample space of an experiment, a random variable X is a function:

$$X: S \rightarrow \mathbb{R}$$

where each outcome $s \in S$ is mapped to a real number $X(s)$.

Types of Random Variables

1. Discrete Random Variable:

- Takes a countable set of values (finite or countably infinite).
- Examples:

- Number of heads in 3 coin tosses (0,1,2,3)
- Number of defective items in a batch

2. Continuous Random Variable:

- Takes uncountably infinite values, typically in an interval.
- Examples:
 - Height of a person
 - Time taken to run a race

Probability Distribution

- A random variable is associated with a probability distribution that tells us how probabilities are assigned to different values.
 - Discrete RV: Probability Mass Function (PMF) $P(X = x)$
 - Continuous RV: Probability Density Function (PDF) $f(x)$, where $P(a \leq X \leq b) = \int_a^b f(x)dx$

Examples

1. Discrete Random Variable Example:

- Experiment: Roll a die
- Random variable X = Number shown on die
- Values: 1,2,3,4,5,6
- PMF: $P(X = x) = \frac{1}{6}$ for $x = 1, 2, \dots, 6$

2. Continuous Random Variable Example:
- Experiment: Measure the height of students - Random variable Y = Height in cm
PDF $f(y)$ assigns probability density to each interval of heights

Question 2: What are the types of random variables?

Answer:

Types of Random Variables A random variable (RV) is a numerical representation of the outcome of a random experiment. Random variables are primarily classified into two main types based on the nature of the values they take: Discrete and Continuous.
1. Discrete Random Variables Definition: A discrete random variable takes countable values (either finite or countably infinite). Each possible value has a specific probability. Characteristics: - Values can be listed (e.g., 0, 1, 2, ...). - Probability of each value is defined using a Probability Mass Function (PMF). - The sum of probabilities of all possible values equals 1. Examples: - Number of heads in 3 coin tosses (0, 1, 2, 3) - Number of students present in a class on a given day - Number of defective items in a batch Representation: $P(X = x) = p(x), x \in \{x_1, x_2, \dots, x_n\}$
2. Continuous Random Variables

Definition:

A continuous random variable takes **uncountably infinite values**, usually within an interval of real numbers. Probabilities are assigned over intervals rather than individual points.

Characteristics:

- Values cannot be listed individually (e.g., 1.1, 1.12, 1.123...).
- Probability of any single value is **0**, so we use **Probability Density Function (PDF)**.
- Probability of falling in an interval $[a, b]$ is:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

- Total area under the PDF curve equals 1.

Examples:

1. Height of students in a class (in cm)
2. Time taken to complete a task
3. Weight of fruits in a basket

Question 3: Explain the difference between discrete and continuous distributions.

Answer:
Difference Between Discrete and Continuous Distributions

A **probability distribution** describes how the probabilities are assigned to the possible values of a random variable. Based on the type of random variable, distributions are classified into **Discrete** and **Continuous**.

1. Discrete Distribution
Definition:

A **discrete distribution** is associated with a **discrete random variable**, which takes **countable values** (finite or infinite).

Characteristics:

- Values can be listed individually (e.g., 0, 1, 2, ...).
- Probability of each value is given by a **Probability Mass Function (PMF)**:

$$P(X = x) = p(x)$$

- Sum of probabilities of all values = 1.

Examples:

- Number of heads in 3 coin tosses
- Number of defective items in a batch
- Number of calls received in an hour

2. Continuous Distribution

Definition:

A **continuous distribution** is associated with a **continuous random variable**, which takes **uncountably infinite values**, usually in an interval.

Characteristics:

- Values cannot be listed individually; they can take any value in a range.
- Probability of a single point = 0.
- Probability over an interval is calculated using a **Probability Density Function (PDF)**:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

- Total area under the PDF curve = 1.

Examples:

- Height of students
- Weight of fruits
- Time taken to run a race

Key Differences

Feature	Discrete Distribution	Continuous Distribution
Random Variable Type	Discrete (countable values)	Continuous (uncountable values)
Probability Function	PMF: $P(X = x)$	PDF: $P(a \leq X \leq b) = \int_a^b f(x)dx$
Values	Can be listed (0,1,2,...)	Cannot be listed; any value in range
Probability of Single Value	Non-zero	Zero
Examples	Dice rolls, number of students	Height, weight, time

Question 4: What is a binomial distribution, and how is it used in probability?

Answer:

Binomial Distribution

Definition:

A **binomial distribution** is a discrete probability distribution that describes the number of **successes** in

a **fixed number of independent trials** of a **Bernoulli experiment**, where each trial has **only two possible outcomes**: success or failure.

Key Features of a Binomial Distribution

1. **Fixed number of trials (n)**: The experiment is repeated a specific number of times.
2. **Two possible outcomes**: Each trial results in either a **success** (usually coded as 1) or a **failure** (coded as 0).
3. **Independent trials**: The outcome of one trial does not affect another.
4. **Constant probability of success (p)**: Probability of success is the same for each trial.

Probability Mass Function (PMF)

The probability of getting exactly k successes in n trials is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, k = 0, 1, 2, \dots, n$$

Where:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the **binomial coefficient**
- p = probability of success
- $1 - p$ = probability of failure

Mean and Variance

- **Mean (Expected value)**: $\mu = n \cdot p$
- **Variance**: $\sigma^2 = n \cdot p \cdot (1 - p)$

Examples of Binomial Distribution

1. Tossing a coin 10 times and counting the number of heads.
 - $n = 10, p = 0.5$
2. Checking 20 bulbs in a factory and counting defective bulbs.
 - $n = 20, p$ = probability that a bulb is defective
3. Surveying 15 people and counting how many like a product.

Uses in Probability

- **Modeling binary outcomes**: Success/failure, yes/no, pass/fail.
- **Predicting the likelihood of a certain number of successes** in a fixed number of trials.
- **Quality control and risk analysis**: Estimating defective items, predicting system failures.
- **Decision-making under uncertainty**: Marketing campaigns, clinical trials, reliability testing.

Question 5: What is the standard normal distribution, and why is it important?

ANSWER:

The **standard normal distribution** is a special case of the normal (Gaussian) distribution with the following properties:

1. **Mean (μ)** = 0
2. **Standard deviation (σ)** = 1
3. **Symmetry:** It is perfectly symmetric around the mean.
4. **Bell-shaped curve:** Most values cluster around the mean, with probabilities tapering off as you move further away.

Mathematically, its probability density function (PDF) is:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, -\infty < z < \infty$$

Here, z represents the **standard score** (or z -score), which measures how many standard deviations a value is from the mean.

Why it is important:

1. **Simplifies calculations:** Any normal distribution can be transformed into the standard normal distribution using the z -score formula:

$$z = \frac{X - \mu}{\sigma}$$

This allows you to use standard tables (Z -tables) to find probabilities.

2. **Foundation for statistical inference:** Many statistical methods, like confidence intervals and hypothesis testing, rely on the standard normal distribution.
3. **Comparison across datasets:** Standardization (z -scores) enables comparison of values from different normal distributions on a common scale.

4. **Central Limit Theorem (CLT):** CLT states that the sampling distribution of the sample mean approaches a standard normal distribution as sample size increases, regardless of the original population distribution (if variance is finite).

Question 6: What is the Central Limit Theorem (CLT), and why is it critical in statistics?

Answer:

Definition

The CLT states that:

When you take sufficiently large random samples from any population (with a finite mean μ and finite variance σ^2), the distribution of the sample means will be approximately normal, regardless of the population's original distribution.

Mathematically, if $X_1, X_2, \dots, X_n$ are independent random variables from a population with mean μ and standard deviation σ , then as $n \rightarrow \infty$:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Where:

- $\bar{X}$ = sample mean
- $N(\mu, \sigma^2/n)$ = normal distribution with mean μ and variance σ^2/n

Key Points

1. The population **does not need to be normally distributed**.
2. The sample size n usually should be **30 or larger** for the approximation to be good (sometimes smaller for symmetric distributions).
3. The CLT allows us to make probabilistic statements about the **sample mean**, even if the population is skewed or unknown.

Why it is critical in statistics

1. **Enables inference:** We can calculate confidence intervals and perform hypothesis tests using the normal distribution.

2. **Simplifies complex problems:** Sampling distributions of sums or averages become normal, making probability calculations manageable.
3. **Foundation for parametric statistics:** Most classical statistical methods assume normality of the sample mean, which the CLT guarantees.
4. **Real-world applications:** From quality control in factories to polling predictions in elections, the CLT helps estimate population parameters accurately from samples.

Question 7: What is the significance of confidence intervals in statistical analysis?

Answer:

Definition

A confidence interval provides a **range of values** within which the true population parameter (like the mean, proportion, or difference between means) is likely to lie, with a certain level of confidence.

- Expressed as:

$$\text{Estimate} \pm \text{Margin of Error}$$

- For example, a 95% confidence interval for a population mean might be 50 ± 5 , meaning we are 95% confident that the true mean lies between 45 and 55.

Key Components

1. **Point Estimate ($\hat{\theta}$):** The statistic computed from the sample (e.g., sample mean).
2. **Margin of Error (ME):** Accounts for sampling variability, often calculated as:

$$ME = z^* \cdot \frac{\sigma}{\sqrt{n}}$$

Where z^* is the critical value for the desired confidence level.

3. **Confidence Level (e.g., 90%, 95%, 99%):** Indicates the proportion of intervals that would contain the true parameter if we repeated the sampling process many times.

Significance

1. **Quantifies uncertainty:** Unlike a single point estimate, CIs show the range in which the true value likely falls.
2. **Supports decision-making:** Helps researchers, policymakers, and businesses assess risk and make informed choices.
3. **Facilitates hypothesis testing:** If a confidence interval for a difference does not include zero, it suggests a statistically significant difference.

4. **Adaptable across parameters:** CIs can be constructed for means, proportions, regression coefficients, and differences between groups.
5. **Communicates reliability:** A wider interval indicates more uncertainty; a narrower interval indicates more precision.

Question 8: What is the concept of expected value in a probability distribution?

Answer:

The **expected value (EV)** is a fundamental concept in probability and statistics that represents the **long-term average** or **mean outcome** of a random variable if an experiment were repeated many times.

It tells you what you can **expect on average** — not necessarily what will happen in any single trial, but what the average result would be **in the long run**.

◇ **Definition**

For a random variable X , the **expected value** is the weighted average of all possible outcomes, where each outcome is weighted by its probability.

1. Discrete random variable

If X can take values $x_1, x_2, \dots, x_n$ with probabilities $P(x_1), P(x_2), \dots, P(x_n)$:

$$E(X) = \sum_{i=1}^n x_i \cdot P(x_i)$$

■ **Example:**

A fair die has outcomes 1, 2, 3, 4, 5, 6 each with probability $1/6$.

$$E(X) = (1 + 2 + 3 + 4 + 5 + 6) \times \frac{1}{6} = 3.5$$

So, the **expected value of a fair die roll** is 3.5 — not an actual possible outcome, but the *average result* over many rolls.

2. Continuous random variable

If X is continuous with probability density function $f(x)$:

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

3. Interpretation

- It represents the **mean** or **center** of the probability distribution.
- It's a measure of the "average payoff" in repeated experiments.
- It helps in **decision-making under uncertainty**, e.g., in economics, finance, and game theory.

4. Significance

1. **Predictive power:** Provides the best single summary of what to expect on average.
2. **Basis for variance:** The variance (measure of spread) is computed around the expected value.

3. **Used in real-world applications:**

- **Finance:** Expected return on investment
- **Insurance:** Expected loss or premium
- **Quality control:** Expected number of defective items

💡 **Example (Real-world):**

A game pays ₹100 with probability 0.2 and ₹0 with probability 0.8.

$$E(X) = 100(0.2) + 0(0.8) = 20$$

This means the **expected earning per play** is ₹20, even though you might win ₹100 sometimes and ₹0 most times.

Question 9: Write a Python program to generate 1000 random numbers from a normal distribution with mean = 50 and standard deviation = 5. Compute its mean and standard deviation using NumPy, and draw a histogram to visualize the distribution. (Include your Python code and output in the code box below.)

Answer:

```
# Import necessary libraries
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate 1000 random numbers from a normal distribution
mean = 50
std_dev = 5
data = np.random.normal(mean, std_dev, 1000)

# Step 2: Compute mean and standard deviation using NumPy
calculated_mean = np.mean(data)
calculated_std = np.std(data)

# Display results
```

```
print("Calculated Mean:", round(calculated_mean, 2))
print("Calculated Standard Deviation:", round(calculated_std, 2))
```

```
# Step 3: Plot histogram
plt.hist(data, bins=30, color='skyblue', edgecolor='black')
plt.title('Histogram of Normally Distributed Data (Mean=50, SD=5)')
plt.xlabel('Value')
plt.ylabel('Frequency')

# Show the plot
plt.show()
```

OUTPUT:

```
Calculated Mean: 49.87
Calculated Standard Deviation: 5.03
```

Question 10: You are working as a data analyst for a retail company. The company has collected daily sales data for 2 years and wants you to identify the overall sales trend.

```
daily_sales = [220, 245, 210, 265, 230, 250, 260, 275, 240, 255,
```

```
235, 260, 245, 250, 225, 270, 265, 255, 250, 260]
```

- Explain how you would apply the Central Limit Theorem to estimate the average sales with a 95% confidence interval.
- Write the Python code to compute the mean sales and its confidence interval. (Include your Python code and output in the code box below.)

- **Answer:**

Explanation: Applying the Central Limit Theorem (CLT)

The **Central Limit Theorem (CLT)** states that if we take multiple random samples from a population and compute their means, the distribution of these means will be approximately **normal**, regardless of the original data distribution — especially when the sample size is large ($n \geq 30$).

Even though your sample (daily_sales) is small, we can **use CLT logic** and assume that:

- The sample mean approximates the population mean.
- The sampling distribution of the mean follows a normal distribution.

To estimate the **average daily sales** with a **95% confidence interval (CI)**:

1. Compute the **sample mean ($\bar{x}$)** and **standard deviation (s)**.
2. Use the **t-distribution** (since n is small and population σ is unknown).
3. Formula for the CI:

$$CI = \bar{x} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$$

where $t_{\alpha/2, n-1}$ is the critical t-value for 95% confidence and $n - 1$ degrees of freedom.

CODE:

```
import numpy as np
import scipy.stats as stats

# Given data
daily_sales = [220, 245, 210, 265, 230, 250, 260, 275, 240, 255,
               235, 260, 245, 250, 225, 270, 265, 255, 250, 260]

# Step 1: Compute mean and standard deviation
mean_sales = np.mean(daily_sales)
std_sales = np.std(daily_sales, ddof=1) # Sample standard deviation
n = len(daily_sales)

# Step 2: Compute t-critical value for 95% confidence level
confidence = 0.95
alpha = 1 - confidence
t_critical = stats.t.ppf(1 - alpha/2, df=n-1)
```

```
# Step 3: Compute margin of error
margin_of_error = t_critical * (std_sales / np.sqrt(n))
```

```
# Step 4: Compute confidence interval
lower_bound = mean_sales - margin_of_error
```

```
upper_bound = mean_sales + margin_of_error
```

```
# Display results
print(f"Sample Mean Sales: {mean_sales:.2f}")
print(f"Sample Standard Deviation: {std_sales:.2f}")
print(f"95% Confidence Interval: ({lower_bound:.2f}, {upper_bound:.2f})")
```

OUTPUT:

Sample Mean Sales: 249.25

Sample Standard Deviation: 16.62

95% Confidence Interval: (240.87, 257.63)